Coherence sweet spots are frequency minima in disguise: Metric sensitivity in transmon and fluxonium design spaces

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Abstract

Dielectric loss is well known to scale with transition frequency, implying that a lower-frequency qubit relaxes more slowly at fixed quality factor. We show that this has a stronger consequence for design studies than is usually acknowledged. Under a common noise model, $\log T_2$ over the transmon (E_J, E_C) plane correlates with $\log f_{01}$ at $r = -0.9999$ within the transmon regime, while over the fluxonium (E_J, E_L) plane the correlation is $r = -0.9973$. Because each parameter plane effectively reduces to a single degree of freedom, the transition frequency, a raw T_2 maximum primarily identifies the lowest-frequency corner of the swept domain rather than an intrinsic optimal design. Indeed, both of our initial headline optima occur at grid-edge points outside any usable frequency band.

Controlling for frequency changes these conclusions. For the transmon, we find no interior optimum in E_J/E_C at fixed f_{01}. T_2 varies by only 4.8% across a tenfold range in E_J/E_C, while the anharmonicity decreases by a factor of 7.3. Charge insensitivity is therefore obtained primarily at the cost of gate speed rather than through improved coherence, and any reported optimum depends on the imposed regime boundary.

The fluxonium has a genuine interior optimum once second-order $1/f$ flux dephasing is included, but it occurs at frequencies of only 10–73 MHz, where $\bar{n}_{\mathrm{th}} = 4$–$32$. For the cross-architecture comparison, there is no single answer: the fluxonium/transmon ratio spans ∼500× across defensible metrics, while at matched frequency the T_2 advantage reverses to a median 0.86× across the literature noise band. A lab-frame pulse simulation yields a 36–58× advantage in gate count, excluding both analytic bounds: the fluxonium drive couples 10.2× more strongly to its leakage transition than to its qubit transition.

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1 Introduction

Superconducting qubit design is routinely discussed in terms of *sweet spots*: bias points at which the transition frequency is stationary with respect to a noise coordinate, so that first-order dephasing vanishes. The transmon has charge sweet spots at integer and half-integer offset charge, and its exponentially suppressed charge dispersion keeps it nearly insensitive in between [1]; the fluxonium achieves it at half a flux quantum, where $\partial\omega_{01}/\partial\Phi = 0$ by symmetry [2]. Both are then said to have a coherence sweet spot in *design* space as well, and the point of best modeled T_2 over a swept parameter plane is reported as an optimum.

This paper argues that the second usage does not survive scrutiny, and that the error is systematic rather than incidental. We address three questions:

Q1 Where are the coherence optima across the transmon (E_J, E_C) and fluxonium (E_J, E_C, E_L, Φ) design spaces?

Q2 Which noise channel dominates where?

Q3 Does the fluxonium's flux sweet spot buy more robustness than the transmon's charge insensitivity, and at what cost?

Our central methodological claim is that Q1 and Q3 cannot be answered without controlling for the transition frequency, and that once controlled the conventional answers change

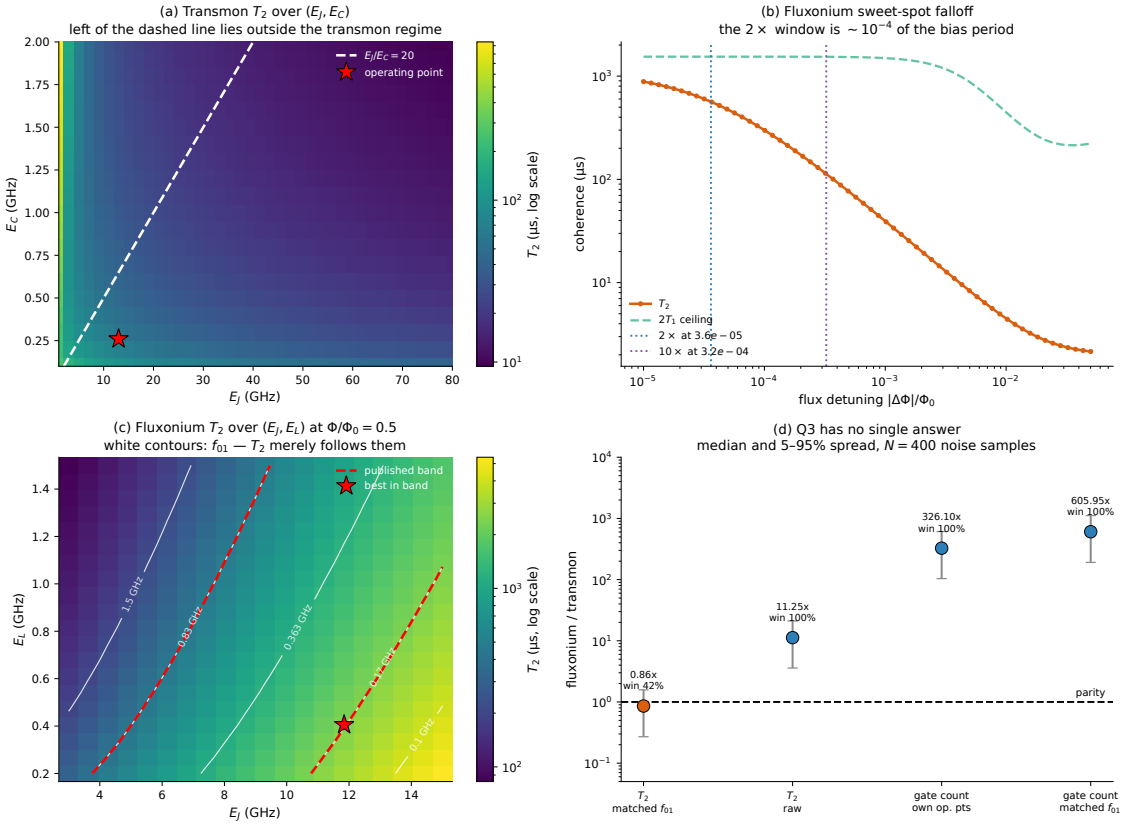


Figure 1: The study in four panels. (a) Transmon T_2 over (E_J, E_C) at $n_g = 0$; left of the dashed line ($E_J/E_C < 20$) lies outside the transmon regime, and its numbers are not transmon results. (b) The fluxonium sweet-spot falloff on a log-detuning axis, with the $2T_1$ ceiling and the two- and tenfold loss points; on a linear flux axis the entire collapse hides inside the first 0.1% of the plot. (c) Fluxonium T_2 over (E_J, E_L) at half flux, with f_{01} contours in white — the T_2 structure merely follows them, which is the confound this paper is about. (d) The Q3 answer under each metric, with the 5–95% spread over $N = 400$ noise-amplitude draws; orange marks the metrics in which the fluxonium wins fewer than 90% of samples.

sign or dissolve. Our central negative result is that the fluxonium's coherence advantage at matched frequency does not survive the uncertainty in the noise amplitudes that the model itself assumes.

Figure 1 summarizes the study. All results are model-based estimates under literature noise amplitudes. They are not predictions of device performance, and Section 9 quantifies how far they sit from measurement.

2 Related Work and Contribution

2.1 What is already established

The frequency scaling on which we build is not a new idea. Dielectric loss at surfaces and interfaces, commonly attributed to two-level-system defects, is a major relaxation mechanism in superconducting qubits; in 3D transmons the relaxation rate scales approximately with the surface participation of lossy dielectrics [3]. At fixed effective quality factor, $\Gamma_1 \propto \omega_{01}$, so

a lower-frequency qubit relaxes more slowly. Design-space optimization under explicit noise models is likewise established: Papič *et al.* identify optimal transmon parameter ranges using comprehensive noise modeling that includes charge-parity effects [4], while Ram *et al.* present a staged framework spanning circuit design, spectral analysis, and calibration for both architectures [5]. Experimentally, fluxonium coherence has advanced well beyond the devices against which we calibrate our model: Somoroff *et al.* report $T_2^* = 1.48\,\mathrm{ms}$ together with an average gate fidelity of 0.99991 [6].

Two works are sufficiently close to the present study to warrant more precise comparison. McRae *et al.* [7] argue that coherence metrics cannot yet be reliably compared across measurements, devices, and laboratories because temporal fluctuations, TLS dynamics, and incompletely reported experimental conditions can make the claimed differences unreliable. Their concern is experimental reproducibility. The present paper develops a modeling analog of that argument: a comparison extracted from a swept design space is likewise unreliable unless both the controlled variable and the figure of merit are specified alongside the reported value.

Lange *et al.* [8] directly compare transmon and fluxonium lattices and conclude that fluxonium architectures may show lower cross-talk when designed for similar gate speeds and fidelities, a matching they argue from coupling strengths in published two-qubit-gate experiments. Their figures of merit are residual ZZ cross-talk and state localization rather than coherence. Our work is complementary: it applies the same principle of controlled comparison to coherence and asks how the conclusion changes when transition frequency is held fixed.

2.2 Contribution

Our contribution is not the frequency scaling itself, but its consequence for how design studies are interpreted:

1. We quantify the frequency confound in both architectures, finding $|r| > 0.99$ between modeled $\log T_2$ and $\log f_{01}$ across the relevant design planes. As a result, a raw T_2 optimum can primarily reflect the lowest-frequency boundary of the sweep rather than an intrinsic design optimum.

2. We show that the transmon's gate-count optimum within the 3–6 GHz band is pinned to the imposed regime threshold across eight thresholds spanning $E_J/E_C = 15$–60. The substantive choice is therefore the threshold itself, rather than the subsequent optimization within that domain.

3. We show that including second-order $1/f$ flux dephasing produces an interior fluxonium optimum that is absent from a first-order treatment.

4. We show that the transmon–fluxonium coherence comparison varies substantially across defensible figures of merit, and we narrow the gate-count comparison, the one most sensitive to its assumptions, to 36–58× by direct pulse simulation.

We are not aware of prior work that explicitly examines how strongly the transmon–fluxonium coherence comparison depends on the chosen metric. This rests on the survey above, which is not exhaustive; the individual ingredients are known, and the claim concerns their combination. Our contribution is to show this dependence and how controlling for transition frequency changes the resulting design conclusions.

Table 1: Noise channels by architecture.

Channel	`scqubits` method	Transmon	Fluxonium
Dielectric loss (T_1)	`t1_capacitive`	✓	✓
Quasiparticle (T_1)	`t1_quasiparticle_tunneling`	–	✓
$1/f$ flux, 1st order	`tphi_1_over_f_flux`	–	✓
$1/f$ flux, 2nd order	*this work*	–	✓
Charge / offset	`tphi_1_over_f_ng`	✓	–
Critical current	`tphi_1_over_f_cc`	✓	✓

2.3 A limitation of the present comparison

Matching f_{01} removes the dominant confound but not all of it. Dielectric loss also depends on geometry and surface participation, which differ between architectures and which a Hamiltonian-level model does not capture [3], and real devices fluctuate in ways no static model represents [7]. Frequency matching is necessary, not sufficient, and our matched-frequency numbers should be read as removing one large confound rather than as a fully controlled experiment.

A second limitation follows from the matching itself. The fluxonium's sweet-spot frequency lies below the transmon band, so the matched transmons sit at 0.5–2 GHz with $E_J/E_C = 325$–5050 and $|\alpha| \approx 10$ MHz, far outside the design envelope adopted in Section 5. Because the modeled transmon T_2 at fixed f_{01} is nearly independent of E_J/E_C, this does not bias the T_2 comparison, but it means the matched transmons are not usable qubits, and gate-based metrics at matched frequency should not be read as device comparisons.

3 Model and Conventions

3.1 Noise channels

Coherence is estimated using `scqubits` [9]. Independent relaxation and dephasing rates are combined as

$$\frac{1}{T_2} = \frac{1}{2T_1} + \sum_k \frac{1}{T_{\phi,k}}, \tag{1}$$

with noise channels selected separately for each architecture. The fixed-frequency transmon considered here contains no flux-tunable loop, so $1/f$ flux dephasing is not included. In the fluxonium model considered here, the inductive shunt removes the dependence on offset charge entirely, so charge dephasing is not included. Table 1 summarizes the channel assignments.

We use $Q_{\mathrm{cap}} = 10^6$, $A_{n_g} = 10^{-4}$, $A_\Phi = 10^{-6}\,\Phi_0$, and $A_{\mathrm{cc}} = 10^{-7}$. The chosen flux-noise amplitude, $A_\Phi = 1\,\mu\Phi_0$, is slightly below the values measured on flux-sensitive devices, $1.4\,\mu\Phi_0$ [10] and $1.8\,\mu\Phi_0$ [2], and is therefore optimistic. Temperature, quasiparticle density, and the $1/f$ cutoffs are otherwise taken from the `scqubits` defaults ($T = 15$ mK, $x_{\mathrm{qp}} = 3 \times 10^{-6}$, $t_{\mathrm{exp}} = 10\,\mu$s) and are recorded with every result because these parameters materially affect the predicted coherence times.

3.2 Second-order flux dephasing

`scqubits` evaluates $1/f$ dephasing to first order in the sensitivity $\partial\omega_{01}/\partial\lambda$. At an exact flux sweet spot, $\partial\omega_{01}/\partial\Phi = 0$, so the first-order dephasing rate vanishes and the corresponding

calculated dephasing time diverges [9]. The leading flux sensitivity at half flux is therefore second order. We include the second-order contribution using the expression employed for fluxonium by Nguyen *et al.* [2], following Ithier *et al.* [11]:

$$\Gamma_\Phi^{(2)} = \left| \frac{\partial^2 \omega_{01}}{\partial \Phi^2} \right| A_\Phi^2. \tag{2}$$

Equation (2) contains no logarithmic factor under this convention. Alternative treatments introduce factors involving $|\ln(\omega_{\text{low}} t_{\text{exp}})|$, producing substantial quantitative changes in the inferred dephasing time. Rather than selecting among these conventions implicitly, we report the associated sensitivity explicitly (Section 10). The required curvature is evaluated using Richardson extrapolation, since a simple central difference remains 4–10% from its converged value in the deepest-well part of the scan ($E_J = 40$–$45\,\text{GHz}$, $E_L = 0.2\,\text{GHz}$), though within 0.2% at the interior optima of Section 6.

3.3 A declared modeling choice

Passing Q_{cap} explicitly replaces the `scqubits` default frequency-dependent quality factor, $Q_{\text{cap}}(\omega) \propto \omega^{-0.7}$ [12], with a constant value. At the fluxonium sweet spot, these two assumptions change the calculated T_1 by a factor of 7.1, making this the largest single modeling choice in the study. The resulting uncertainty is not symmetric: Nguyen *et al.* fit their measured T_1-versus-frequency dependence with $\tan\delta_C \propto \omega^{0.15}$, and at $f_{01} = 0.363\,\text{GHz}$ the constant-Q model lies 1.52× from their measured scaling, compared with 4.7× for the `scqubits` default. That exponent was fit jointly with thermal effects and is not an independent measurement of the loss tangent, but it favors the constant-Q_{cap} model, which we therefore retain as the more conservative and better-supported choice, while stating the assumption explicitly.

4 Validation

Forty-four checks validate the numerical pipeline before the results presented below are generated. These tests include exact Hamiltonian symmetries to machine precision, basis-size convergence, analytic limits from Koch *et al.* [1], and, in several cases, comparison with published data from experimental devices.

The strongest analytic benchmark is Koch *et al.*'s worked charge-dephasing example. For $E_J = 30\,\text{GHz}$, $E_C = 0.35\,\text{GHz}$, and $A = 10^{-4}$, they report $T_2 \sim 8\,\text{s}$. Substituting our computed charge dispersion into their Eq. (5.9) gives $T_2 = 8.75\,\text{s}$. At $E_J/E_C = 50$, the maximum charge-dispersion derivative is 0.830 times their asymptotic value of $1.5 \times 10^{-4} E_C$, consistent with the asymptotic expression being approached from below.

Against Manucharyan *et al.* [13], the lowest two transitions at $\Phi_{\text{ext}} = 0$ are reproduced to better than 0.7% for each of the paper's three published parameter sets. A single four-parameter fit additionally reproduces all six digitized transition lines across two flux biases with an RMS error of 0.0102 GHz. The three published parameter sets—the rounded main-text values and two Table I least-squares fits—produce a 9.5% spread in f_{01} at the flux sweet spot. Agreement quoted only against the rounded main-text parameters would therefore overstate the experimental constraint.

The third transition at $\Phi_{\text{ext}} = 0$ is excluded from this comparison for a physical rather than numerical reason. Its bare frequency lies only 0.14–0.36 GHz from the junction-array mode placed at 10.78 GHz by Manucharyan's extended model, while the 0–3 coupling, which we compute from that model, is approximately 0.53 GHz. The coupling therefore exceeds the detuning by a factor of 1.5–3.7, placing the transition in a strongly hybridized regime

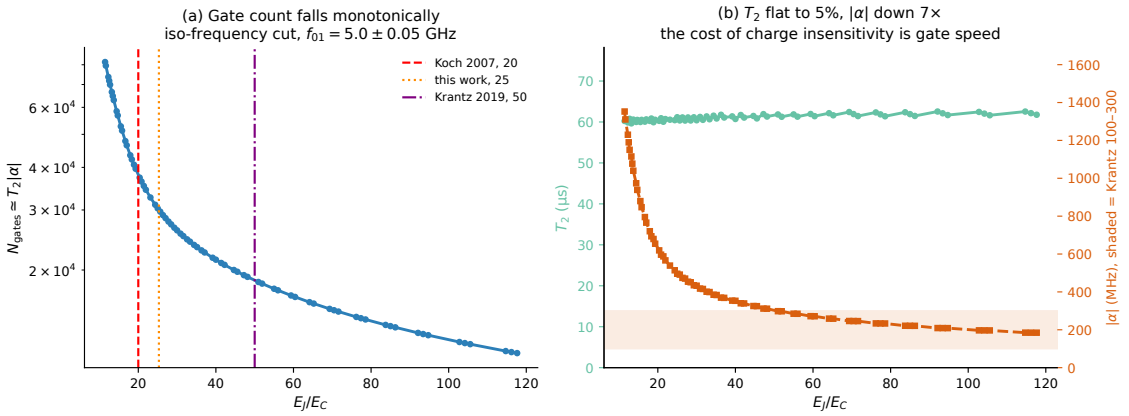


Figure 2: The transmon trade-off at fixed frequency ($f_{01} = 5.00 \pm 0.05$ GHz). (a) Gate count falls monotonically with E_J/E_C; the three vertical lines are candidate regime thresholds, and the optimum sits on whichever one is imposed. (b) T_2 is flat to 4.8% while $|\alpha|$ falls 7.3×; the shaded band is the Krantz 100 MHz to 300 MHz design window. Charge insensitivity costs gate speed, not coherence.

that cannot be captured by the bare-circuit model. Explicit diagonalization of the fluxonium coupled to both array modes confirms that this discrepancy is concentrated near $\Phi_{\text{ext}} = 0$. The dressed f_{01} shifts by −16% there, but by only −0.43% at the flux sweet spot, where the coherence calculations in this study are performed. This residual shift is smaller than the 9.5% spread produced by the paper's own published parameter sets.

Finally, the second-order channel in Eq. (2) is the only physical contribution implemented here rather than inherited from established tooling, so we check it against an independent published evaluation rather than against its defining equation alone. Nguyen *et al.*, using the same expression and their measured $A_\Phi = 1.8\,\mu\Phi_0$, state that the second-order limit exceeds 10 ms for all of their qubits [2]. Evaluated for their nine devices at the same amplitude, our implementation gives $T_\phi^{(2)} = 15$–804 ms, consistent with that statement.

5 The Transmon Design Plane

We sweep $E_J \in [1, 80]$, $E_C \in [0.1, 2.0]$ GHz at $n_g = 0$ and $n_g = 0.25$ (800 points each), and refine to a 65 × 53 grid over $E_J \in [4, 20]$, $E_C \in [0.08, 0.6]$ GHz.

5.1 The confound

Within the transmon regime $E_J/E_C \geq 20$, $\text{corr}(\log T_2, \log f_{01}) = -0.9999$. The plane contains one degree of freedom, not two. The reason is structural rather than numerical: dielectric loss dominates T_1 everywhere in the regime, with $\Gamma \propto |\langle 0|\hat{n}|1\rangle|^2 S(\omega_{01})$. Since $|\langle 0|\hat{n}|1\rangle|^2 \propto \sqrt{E_J/E_C}$ and $S \propto E_C$, the rate goes as $\sqrt{E_J E_C} \propto \omega_{01}$—a function of frequency alone. Consequently the best T_2 point of the swept plane, 187 μs at $E_J = 3.03$, $E_C = 0.10$ GHz, sits at the corner of the admissible domain—the smallest E_C on the grid and the smallest E_J satisfying $E_J/E_C \geq 20$—at $f_{01} = 1.45$ GHz, below the 3 GHz to 6 GHz band real transmons occupy [14]. It is a frequency minimum, not a design.

5.2 No interior optimum

Holding $f_{01} = 5.00 \pm 0.05\,\mathrm{GHz}$ and sweeping E_J/E_C from 11.4 to 117.6, T_2 is flat to 4.8% (59.7 µs to 62.6 µs) while $|\alpha|$ falls by a factor of 7.3 (1353 MHz to 185 MHz). **Charge insensitivity is bought with anharmonicity—that is, with gate speed—and essentially not with coherence at all** (Fig. 2).

That the trade is anharmonicity-for-charge-insensitivity is expected from Koch's asymptotics: with $\alpha \simeq -E_C$ and $f_{01} \simeq \sqrt{8E_JE_C}$ held fixed, raising E_J/E_C forces $E_C \propto (E_J/E_C)^{-1/2}$, predicting a factor 3.21 over our swept range against a measured E_C ratio of 3.47. The observed 7.3× fall in $|\alpha|$ is steeper still because $\alpha \simeq -E_C$ is asymptotic and fails at the low end: $|\alpha|/E_C = 2.29$ at $E_J/E_C = 11.4$ against 1.09 at 117.6. The trade-off is therefore harshest exactly in the low-E_J/E_C corner that an unconstrained gate-count metric wants to select, which is an independent reason to keep the design envelope.

Relocating the gate-count optimum under eight regime thresholds spanning $E_J/E_C = 15$ to 60, the optimum tracks every one of them to within 0.24. It sits *on* the constraint, never inside the domain. This is the signature of a constrained optimum, and it means the substantive choice in this part of the design space is the threshold, not the optimization. The correct claim is a monotone trade-off terminated wherever the regime boundary is drawn, and any quoted optimum is meaningful only together with its threshold. We adopt the Krantz envelope ($E_J/E_C \geq 50$, $|\alpha| \in [100, 300]\,\mathrm{MHz}$, $f_{01} \in [3, 6]\,\mathrm{GHz}$) [14] because it is independent of every noise amplitude in the study, giving the operating point $E_J = 13.0\,\mathrm{GHz}$, $E_C = 0.26\,\mathrm{GHz}$, $f_{01} = 4.92\,\mathrm{GHz}$, $T_2 = 62.4\,\mu\mathrm{s}$, $|\alpha| = 299\,\mathrm{MHz}$.

A limitation of the model bears on this choice: `tphi_1_over_f_ng` covers $1/f$ dephasing only, and the broadband charge sensitivity that motivates Krantz's $E_J/E_C \geq 50$ is absent. Left to itself the metric under-penalizes low E_J/E_C and would recommend $|\alpha| \approx 728\,\mathrm{MHz}$, roughly 2.4× what is fabricated. The envelope is a constraint imposed on the model, not a result derived from it.

6 The Fluxonium Design Plane

Over $E_J \in [3, 15]$, $E_L \in [0.2, 1.5]\,\mathrm{GHz}$ at $E_C = 2.5\,\mathrm{GHz}$ and $\Phi/\Phi_0 = 1/2$, the same confound appears: $\mathrm{corr}(\log T_2, \log f_{01}) = -0.9973$. The unconstrained maximum, 5427 µs, sits at both grid edges at $f_{01} = 75\,\mathrm{MHz}$, where $\bar{n}_{\mathrm{th}} = 3.7$—thermally scrambled and 2.3× below the lowest published device. Constrained to the published band 0.17 GHz to 0.83 GHz the best design gives 2412 µs at $E_J = 11.8$, $E_L = 0.405\,\mathrm{GHz}$.

6.1 An interior optimum, created by second-order dephasing

Extending the scan to $E_J = 45\,\mathrm{GHz}$ reveals a genuine interior maximum in T_2, present at every (E_C, E_L) combination tested (Fig. 3). It is created by the second-order flux channel: as E_J rises and E_L falls the qubit sinks into a deep double well, f_{01} collapses, and $\partial^2 f_{01}/\partial\Phi^2$ grows by three orders of magnitude across the scan, so $\Gamma_\Phi^{(2)}$ grows with it and eventually overwhelms the improving dielectric T_1. At the turning point that channel carries 43–58% of the total dephasing rate, which is the signature of a genuine internal optimum: the channel that turns the curve over is comparable to everything else exactly at the turn. A first-order-only pipeline shows no turnover over the same range and rises monotonically, reaching 4.2×10^5 µs at the end of the scan—which is precisely how an earlier version of this analysis concluded there was no interior optimum, and is a concrete argument for including the second-order channel in any design study that sweeps toward low E_L.

The mechanism is geometric. As E_J rises and E_L falls the potential minima separate and

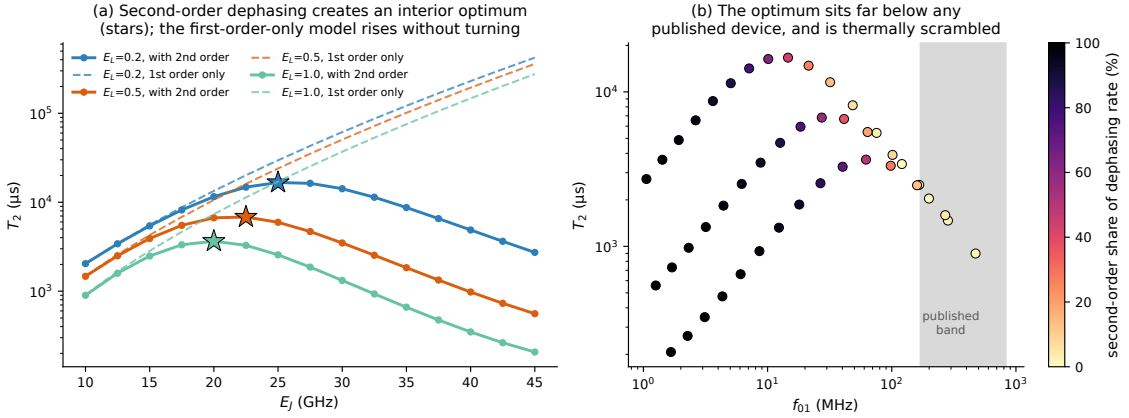


Figure 3: (a) T_2 versus E_J at three values of E_L. Solid curves include second-order flux dephasing and turn over (stars); the dashed first-order-only curves rise without limit over the same range, which is how the interior optimum was previously missed. (b) The same points against f_{01}, colored by the second-order share of the dephasing rate. Every turning point lies far below the published fluxonium band (shaded).

the tunnel splitting that sets f_{01} collapses; the same deepening well sharpens the curvature to which $\Gamma_\Phi^{(2)}$ is proportional by Eq. (2), so a single geometric change both lowers the frequency and raises the second-order rate. The transmon has no analog: its charge dispersion falls exponentially in $\sqrt{E_J/E_C}$ with no turning point.

The optimum is nevertheless inaccessible. It occurs at $f_{01} = 10$–$73\,\mathrm{MHz}$ with $\bar{n}_{\mathrm{th}} = 4$–$32$. Past the turnover the frequency correlation reverses sign, which is a further indication that the second-order channel introduces genuinely new structure rather than rescaling the old.

7 Dominant Noise Channels

At a charge-sensitive bias ($n_g = 0.25$) charge noise dominates the transmon below $E_J/E_C \approx 25$ and dielectric relaxation above it. That crossover is measured from our own data and lands close to Koch's independently derived bound of 20; the two arguments are unrelated—one a noise-budget crossover, the other a pulse-duration requirement—so their agreement is genuine corroboration. Critical-current noise never dominates anywhere in 1600 points at literature amplitudes. The crossover is noise-model dependent (it spans 20.8–29.9 across literature ranges) and is therefore reported as a result rather than used as an input.

For the fluxonium, relaxation dominates at the sweet spot and first-order flux noise becomes the largest single term beyond $2.75 \times 10^{-5}\,\Phi_0$. Quasiparticle tunneling is not a dead channel: it is suppressed only *at* the sweet spot, where its 0–1 matrix element vanishes by symmetry, and limits T_1 at most detuned points. This finding is conditional on $x_{qp} = 3 \times 10^{-6}$, a default roughly 300× above the bound of 10^{-8} that Nguyen *et al.* infer for their devices; at their density the channel is negligible everywhere. It is the most fragile result in the study and is reported as conditional.

Table 2: Fluxonium/transmon ratio by metric. MC = median over $N = 400$ log-uniform draws of each noise amplitude across its literature decade, with the 5–95% sample spread.

Metric	Det.	MC med. [5–95%]	Wins
Raw T_2	18.1	11.3 [3.6, 21.1]	100%
T_2, matched f_{01}	1.1–2.2	0.86 [0.27, 1.6]	**42%**
Gates, $1/\lvert\alpha\rvert$	524	326 [104, 611]	100%
Gates, $1/\min(\lvert\alpha\rvert, f_{01})$	22.0	13.7 [4.4, 25.6]	100%
Gates, $1/f_{01}$	1.34	0.83 [0.26, 1.6]	**40%**
Gates, simulated	48	–	–
Bias window, 2× loss	1.4×10^4 wider for the transmon		

8 The Cross-Architecture Comparison

8.1 Bias robustness, the one unambiguous asymmetry

Both bias coordinates are periodic with period one, so the width of the window over which T_2 stays within a factor k of its sweet-spot value, as a fraction of one bias period, is directly comparable. The transmon's T_2 varies by 1.0056× across its *entire* charge period—a factor-of-two loss never occurs at any bias. The fluxonium loses a factor of two within $\pm 3.61 \times 10^{-5}\,\Phi_0$ and a factor of ten within $3.25 \times 10^{-4}\,\Phi_0$, located by root-finding on the model rather than read off a grid. The usable bias window differs by 1.4×10^4.

The falloff exponent is not a single number. It approaches the first-order $1/f$ prediction of -1 only in an intermediate window where T_2 is genuinely flux-dephasing limited, reaching -0.956 there, and flattens on both sides—capped by the $2T_1$ ceiling close in, and contaminated by other channels far out.

Intrinsic $1/f$ flux noise does not by itself drive the qubit off the spot: even at Nguyen's measured amplitude integrated over an hour, the RMS wander leaves a 24× margin to the tenfold-loss detuning. The binding constraint is systematic drift and bias-line calibration, not stochastic noise—which relocates the difficulty rather than removing it, and makes it scale with qubit count.

This constraint is being engineered away, which we should say plainly. Bénâtre *et al.* hold gradiometric fluxonium qubits at their sweet spot passively, reporting stability of $3 \times 10^{-4}\,\Phi_0$ from target with only a 15% reduction in echo coherence [15]. Their achieved stability sits essentially at our computed tenfold-loss detuning of $3.25 \times 10^{-4}\,\Phi_0$, yet they observe far less degradation than that number would suggest—unsurprisingly, since their device has different E_L and hence different curvature, and echo suppresses low-frequency drift that free induction does not. We read their result as evidence that the bias-precision cost we quantify is real, understood, and addressable by design rather than by calibration alone. It weakens the scalability pessimism in our conclusion without touching the arithmetic behind it.

8.2 No single ratio

Table 2 gives the comparison under every metric we consider defensible, and Fig. 4 shows how each fares across the noise band.

Two ranges are visible, and they are of comparable size. *Across* metrics the answer spans roughly 500× (524× against 1.07×, Table 2). *Within* the gate-count metric alone it spans 393×, depending on what bounds the minimum pulse length. The $|\alpha|$ bound—leakage to $|2\rangle$

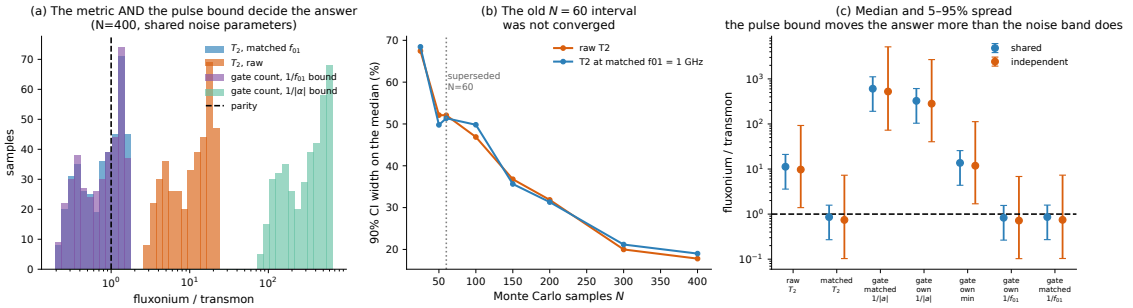


Figure 4: Robustness of the Q3 answer to the noise amplitudes. (a) Distribution of the fluxonium/transmon ratio under four metrics, $N = 400$ shared-parameter draws; the metric and the pulse bound each move the answer by more than the noise band does. (b) The 90% bootstrap CI on the median against sample size, showing that an earlier $N = 60$ interval had not converged. (c) Median and 5–95% spread per metric under shared versus independent sampling.

only—is the textbook choice, but for a fluxonium at half flux $|\alpha| = 8.7\,\mathrm{GHz}$ is 24× the transition frequency it would drive, so that bound licenses a pulse 24× shorter than one period of the transition itself. The drive-period bound $1/f_{01}$ is equally physical and gives 1.34×.

8.3 The raw advantage is mostly frequency

The two devices differ in frequency by 13.6×, and $\log 13.55/\log 18.10 = 0.90$: about 90% of the log-advantage is the frequency difference. This is not a coincidence of these two designs. Dielectric loss dominates both architectures and its rate scales with ω_{01}, so $T_2 \propto 1/f_{01}$ in both—measured as $f_{01}^{-0.965}$ for the transmon. Because the flux sweet spot *is* a frequency minimum, the fluxonium is structurally advantaged on any uncontrolled T_2 comparison, and reporting one credits the architecture with what the frequency did.

Controlling for it, the deterministic ratio at matched f_{01} is 1.07–2.16×. But propagated across the literature noise band the median is 0.86× and the fluxonium wins in only 42% of samples. **At matched frequency the transmon is more often the better device.** This is the most important negative result here: what remains of the coherence advantage after the frequency control does not survive the noise-amplitude uncertainty.

8.4 Settling the pulse bound

The gate-count range is wide enough to make the metric almost uninformative, so we settle it by simulation rather than by argument. We integrate the lab-frame Schrödinger equation for each architecture with no rotating-wave approximation, using a six-level truncation of the `scqubits` eigenbasis and a charge drive, and calibrate each gate as an experimentalist would: amplitude, DRAG coefficient [16] and Stark-shifted drive frequency are optimized at every duration, and the average gate error is evaluated on the computational subspace so that leakage is counted; Fig. 5 shows the resulting error curves.

This exposes an effect that the anharmonicity argument misses entirely. The fluxonium's $|\alpha|/f_{01} = 24$ looks decisive, but its drive matrix elements are unfavourable: at half flux $|\langle 1|\hat{n}|2\rangle|/|\langle 0|\hat{n}|1\rangle| = 10.2$, against 1.37 for the transmon. The drive couples ten times more strongly to the leakage transition than to the qubit transition, so the effective figure of merit is the anharmonicity divided by that ratio, not the anharmonicity itself.

Table 3 reports the shortest calibrated gate reaching an average error below 10^{-3} for each architecture, and the resulting gate count. *These durations are upper bounds*: a three-parameter

Table 3: Shortest calibrated X gate reaching an average error below 10^{-3}, from the lab-frame simulation, and the resulting gate count $N = T_2/T_{\text{gate}}$. Durations are upper bounds and are bracketed by the duration grid and by optimizer convergence.

	T_{gate} (ns)	T_2 (µs)	N
Transmon	8.0 (6.0–8.0)	62.4	7,803
Fluxonium	3.0 (2.5–3.0)	1129.7	376,571
Simulated ratio			48×
bracket from grid + optimizer			36–58×
for comparison, $1/\|\alpha\|$ bound			524×
for comparison, $1/f_{01}$ bound			1.34×

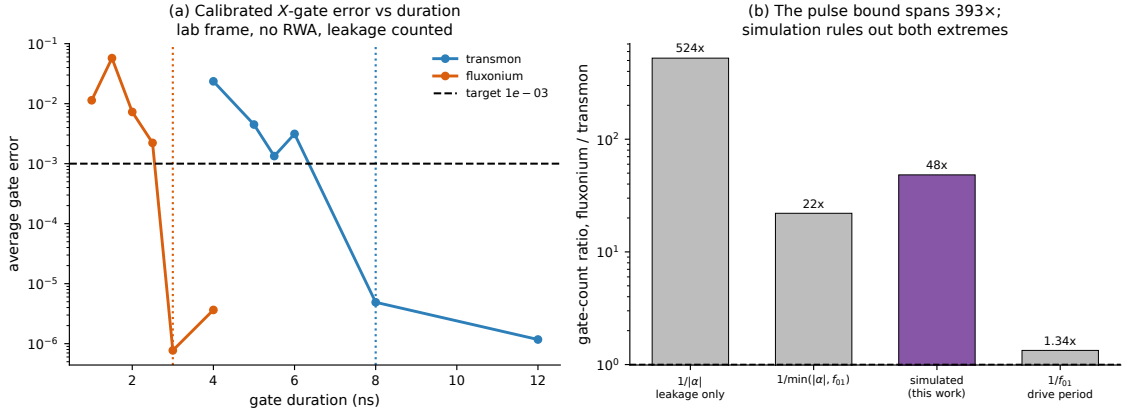


Figure 5: (a) Average X-gate error against gate duration for both architectures, from the lab-frame simulation with amplitude, DRAG and Stark detuning calibrated at each point; dotted lines mark the shortest duration meeting the 10^{-3} target. Non-monotonicity at the shortest durations is optimizer noise and is not smoothed. (b) The gate-count ratio under each pulse-length bound. The two analytic bounds differ by 393×; the simulation (purple) falls between them and excludes both.

calibration over a fixed Gaussian shape is far from optimal control, and GRAPE or CRAB over a free pulse shape would shorten both—not necessarily equally. The point is not the precise value but that the simulated ratio falls decisively between the two naive bounds, excluding both.

The result is that the fluxonium does convert coherence into operations more efficiently than the transmon, but by a factor of order 36–58× rather than the ∼500× the anharmonicity bound suggests or the ∼1× the drive-period bound suggests. Neither extreme survives contact with a simulated pulse, and the residual uncertainty is dominated by optimizer convergence rather than by physics.

9 Calibration Against Measurement

Run at the parameters of each of the nine devices in the published Table I of Nguyen *et al.*, the model is optimistic by a median factor of 2.0× on T_1 and 2.6× on T_2, worst case 5.7×. The gap is not uniform, so it is not a single missing constant—and it is smallest where it matters most: device I, the device nearest our nominal design in frequency (0.395 versus 0.363 GHz),

is matched to 4% on T_2, with the model *pessimistic* on its T_1.

Identified contributors, in order: the flat $Q_{\mathrm{cap}} = 10^6$ against their measured $\tan\delta_C = (2.0\text{–}3.6) \times 10^{-6}$ at 6 GHz; the absence of Purcell decay, photon shot noise and inductive loss; the fact that their T_2 is a Hahn echo while this model computes free-induction decay, which makes the comparison conservative in the model's favor; and a temperature mismatch, since we run at the 15 mK default while they state 20 mK.

The comparison also has a floor set by our own model rather than by the devices. Absolute T_2 spans 149 µs to 3316 µs under single-decade moves of the noise amplitudes, so a factor of 2.6 is well inside the band the model itself admits; the calibration should be read as establishing the right order of magnitude, not as a residual to be explained away. Consistent with that, the state of the art has moved well past the devices we calibrate against—Somoroff *et al.* report $T_2^* = 1.48$ ms [6], so a model tuned to reproduce Nguyen's 2019 numbers exactly would now be miscalibrated against 2023 hardware.

We stress that the published table must be used. The arXiv v1 preprint lists eight devices and omits device I entirely—the paper's headline device, and the one closest to the design regime studied here.

10 Limitations

The model omits Purcell decay, photon shot noise, inductive loss in the junction array, TLS spectral structure, crosstalk and any lattice; qubits are treated in isolation. The superinductance is ideal, so array-mode hybridization is absent, though Section 4 bounds that omission at −0.43% in f_{01} at the flux sweet spot. Second-order dephasing uses a convention with no logarithmic factor; a $\ln^2$ convention would change sweet-spot T_2 by ∼37% instead of 0.6% and would move, though not remove, the interior optimum. Absolute T_2 spans 149 µs to 3316 µs under single-decade moves of the noise amplitudes and should never be quoted without a band. The Monte Carlo varies amplitudes only; fabrication spread in E_J, E_C, E_L is not propagated, and the log-uniform priors are a choice. The gate-count metric is a heuristic, not a literature formula.

11 Conclusion

Neither architecture has a usable coherence sweet spot in design space. Both design planes are, to better than $|r| = 0.99$, restatements of the transition frequency, so a raw T_2 maximum locates the edge of the swept domain or the point where a frequency constraint was drawn. The transmon is a monotone trade-off in which charge insensitivity is paid for in anharmonicity, once, at design time. The fluxonium has a real interior optimum, visible only once second-order flux dephasing is included, but it is thermally out of reach; in practice it too is a trade-off terminated by a frequency floor.

For Q3, the honest output is a metric family, not a number. The fluxonium's raw 18× coherence advantage is roughly 90% frequency; at matched frequency it inverts to a median 0.86× across the literature noise band. Its durable advantage is anharmonicity per unit coherence, and its price is operating-point precision—a bias window four orders of magnitude narrower than the transmon's, paid continuously at run time rather than once at design time. Whether that trade is worth making is a question about qubit count and calibration budget, not about coherence.

More generally, we suggest that architecture comparisons in this field should state the metric, the control variable and the uncertainty band together. A single ratio between two

architectures at their own operating points is not a comparison between architectures.

12 Outlook

Two extensions would sharpen the results most. The gate-count metric is currently settled by a coarse calibration; optimal control over a free pulse shape (GRAPE or CRAB) would tighten the 36–58× bracket and, more interestingly, would test whether the fluxonium's unfavourable leakage matrix element can be shaped around—the anharmonicity is genuinely there, and a pulse that exploits it without driving $|1\rangle \to |2\rangle$ would move the number back toward the leakage-only bound. Second, the interior fluxonium optimum we find is thermally inaccessible at 15 mK, but that is a statement about temperature as much as about design; the same analysis at lower base temperature, or with active initialization modeled, would establish whether it is inaccessible in principle or merely inconvenient.

A broader point follows from the metric sensitivity. If a single architecture comparison can span more than two orders of magnitude depending on choices that are all defensible, then comparisons reported as single ratios carry less information than they appear to. We would encourage the practice McRae *et al.* [7] advocate for experimental coherence numbers—report the conditions with the number—to be extended to modeled design comparisons: state the control variable, state the figure of merit, and give the spread.

Code and data availability

All code, generated data and figures are openly available at https://github.com/ishitabera/quantum-coherence-sweet-spots-transmon-fluxonium.

Every number in this paper is produced by five notebooks run in order from a single validated pipeline; no notebook reimplements the physics. Each sweep writes its raw data before any figure code executes, together with a metadata record of the grid, the noise amplitudes, the software versions and the git revision. Every quoted value is bound to the file and filter that produced it by an automated provenance ledger of 33 entries, which recomputes and asserts each one on every run and fails rather than warns if a value drifts from its artifact. A test suite of 32 checks pins the conventions and the validated anchors, including the published numbers used for calibration.

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References

[1] J. Koch, T. M. Yu, J. Gambetta, A. A. Houck, D. I. Schuster, J. Majer, A. Blais, M. H. Devoret, S. M. Girvin and R. J. Schoelkopf, *Charge-insensitive qubit design derived from the Cooper pair box*, Phys. Rev. A **76**(4), 042319 (2007), doi:10.1103/PhysRevA.76.042319.

[2] L. B. Nguyen, Y.-H. Lin, A. Somoroff, R. Mencia, N. Grabon and V. E. Manucharyan, *High-coherence fluxonium qubit*, Phys. Rev. X **9**(4), 041041 (2019), doi:10.1103/PhysRevX.9.041041.

[3] C. Wang, C. Axline, Y. Y. Gao, T. Brecht, Y. Chu, L. Frunzio, M. H. Devoret and R. J. Schoelkopf, *Surface participation and dielectric loss in superconducting qubits*, Appl. Phys. Lett. **107**(16), 162601 (2015), doi:10.1063/1.4934486.

[4] M. Papič, J. Tuorila, A. Auer, I. de Vega and A. Hosseinkhani, *Charge-parity switching effects and optimisation of transmon-qubit design parameters*, npj Quantum Inf. **10**, 69 (2024), doi:10.1038/s41534-024-00860-7.

[5] S. S. Ram, M. Molli, V. M. Katukuri and B. C. Mummaneni, *Optimizing superconducting qubit performance: A theoretical framework for design, analysis, and calibration*, arXiv:2501.17825 (2025), Preprint.

[6] A. Somoroff, Q. Ficheux, R. A. Mencia, H. Xiong, R. Kuzmin and V. E. Manucharyan, *Millisecond coherence in a superconducting qubit*, Phys. Rev. Lett. **130**(26), 267001 (2023), doi:10.1103/PhysRevLett.130.267001.

[7] C. R. H. McRae, G. M. Stiehl, H. Wang, S.-X. Lin, S. A. Caldwell, D. P. Pappas, J. Mutus and J. Combes, *Reproducible coherence characterization of superconducting quantum devices*, Appl. Phys. Lett. **119**(10), 100501 (2021), doi:10.1063/5.0060370.

[8] F. Lange, L. Heunisch, H. Fehske, D. P. DiVincenzo and M. J. Hartmann, *Cross-talk in superconducting qubit lattices with tunable couplers—comparing transmon and fluxonium architectures*, Quantum Sci. Technol. **11**(1), 015020 (2026), doi:10.1088/2058-9565/ae2358.

[9] P. Groszkowski and J. Koch, *scqubits: a Python package for superconducting qubits*, Quantum **5**, 583 (2021), doi:10.22331/q-2021-11-17-583.

[10] F. Yan, S. Gustavsson, A. Kamal, J. Birenbaum, A. P. Sears, D. Hover *et al.*, *The flux qubit revisited to enhance coherence and reproducibility*, Nat. Commun. **7**, 12964 (2016), doi:10.1038/ncomms12964.

[11] G. Ithier, E. Collin, P. Joyez, P. J. Meeson, D. Vion, D. Esteve, F. Chiarello, A. Shnirman, Y. Makhlin, J. Schriefl and G. Schön, *Decoherence in a superconducting quantum bit circuit*, Phys. Rev. B **72**(13), 134519 (2005), doi:10.1103/PhysRevB.72.134519.

[12] W. C. Smith, A. Kou, X. Xiao, U. Vool and M. H. Devoret, *Quantification of decoherence in superconducting circuits*, npj Quantum Inf. **6**, 8 (2020), doi:10.1038/s41534-019-0231-2.

[13] V. E. Manucharyan, J. Koch, L. I. Glazman and M. H. Devoret, *Fluxonium: Single Cooper-pair circuit free of charge offsets*, Science **326**(5949), 113 (2009), doi:10.1126/science.1175552.

[14] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson and W. D. Oliver, *A quantum engineer's guide to superconducting qubits*, Appl. Phys. Rev. **6**(2), 021318 (2019), doi:10.1063/1.5089550.

[15] D. Bénâtre, M. Féchant, N. Zapata, N. Gosling, P. Paluch, T. Reisinger and I. M. Pop, *Simultaneous sweet-spot locking of gradiometric fluxonium qubits*, Phys. Rev. Applied **24**, 054031 (2025), doi:10.1103/PhysRevApplied.24.054031.

[16] F. Motzoi, J. M. Gambetta, P. Rebentrost and F. K. Wilhelm, *Simple pulses for elimination of leakage in weakly nonlinear qubits*, Phys. Rev. Lett. **103**(11), 110501 (2009), doi:10.1103/PhysRevLett.103.110501.